WIRESHARK PACKET ANALYSIS REPORT

Reconnaissance & Packet Capture — Task 3

Alfido Tech Internship

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Tool Used	Wireshark 4.6.4 + Npcap 1.86
Capture File	task3_capture.pcapng
Task	Task 3 — Wireshark Packet Analysis

1. Introduction

This report documents the findings of a live network packet capture and analysis exercise performed using Wireshark on a Windows 11 machine. The objective was to capture real-time network traffic, identify protocols (DNS, TCP, TLS), analyze packet structure and flags, and detect any unusual or malformed packets. All traffic captured is from the analyst's own network interface.

2. Scope & Methodology

Interface	Wi-Fi (192.168.31.139)
Capture Duration	~86 seconds
Total Packets	9,815
Default Gateway	192.168.31.1
OS	Windows 11

- Phase 1: Live packet capture on Wi-Fi interface using Wireshark
- Phase 2: Protocol filtering — DNS, TCP SYN, TLS, ARP, tcp.analysis.flags
- Phase 3: Analysis, anomaly detection, and report preparation

3. Protocols Identified

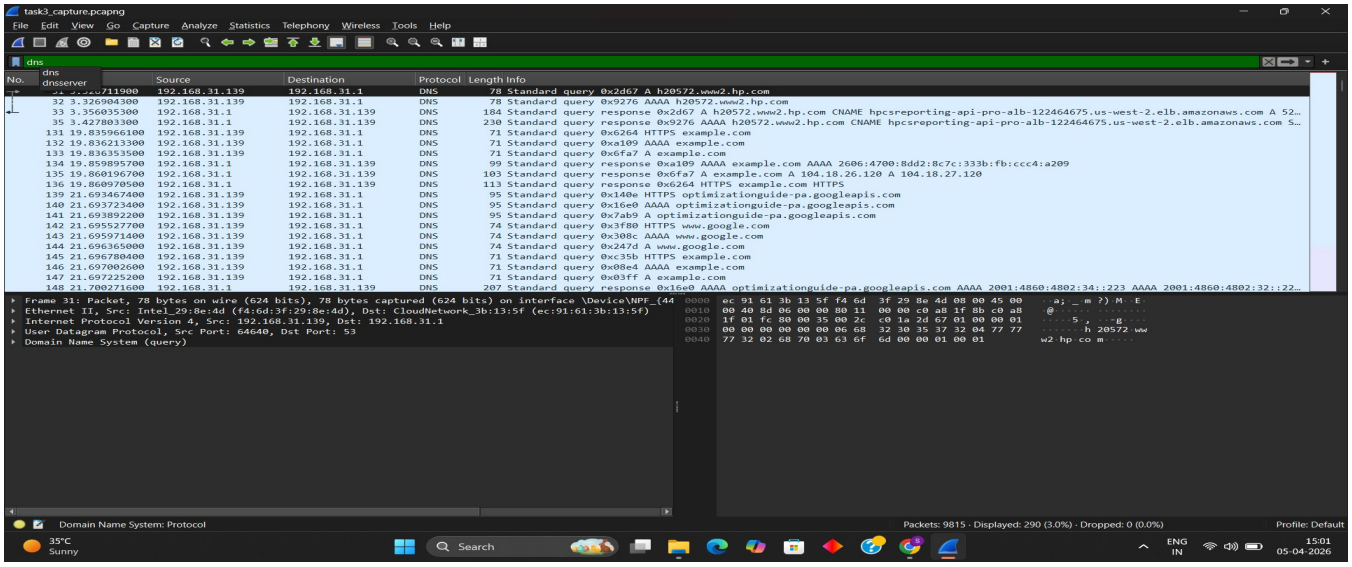
Protocol	Filter Used	Packets	Description
DNS	dns	290 (3.0%)	Domain name queries to google.com, example.com, googleapis.com
TCP SYN	tcp.flags.syn==1 && tcp.flags.ack==0	56 (0.6%)	TCP 3-way handshake initiations to port 443 & 80
TLS 1.2/1.3	tls	781 (8.0%)	Encrypted HTTPS application data, Client Hello observed

ARP	arp	6 (0.1%)	Address Resolution Protocol — gateway MAC discovery
QUIC	(in all packets)	200+	Google/YouTube QUIC protocol traffic
TCP Anomalies	tcp.analysis.flags	135 (1.4%)	Retransmissions, ZeroWindow, Out-of-Order, Dup ACK

4. Wireshark Packet Analysis Results

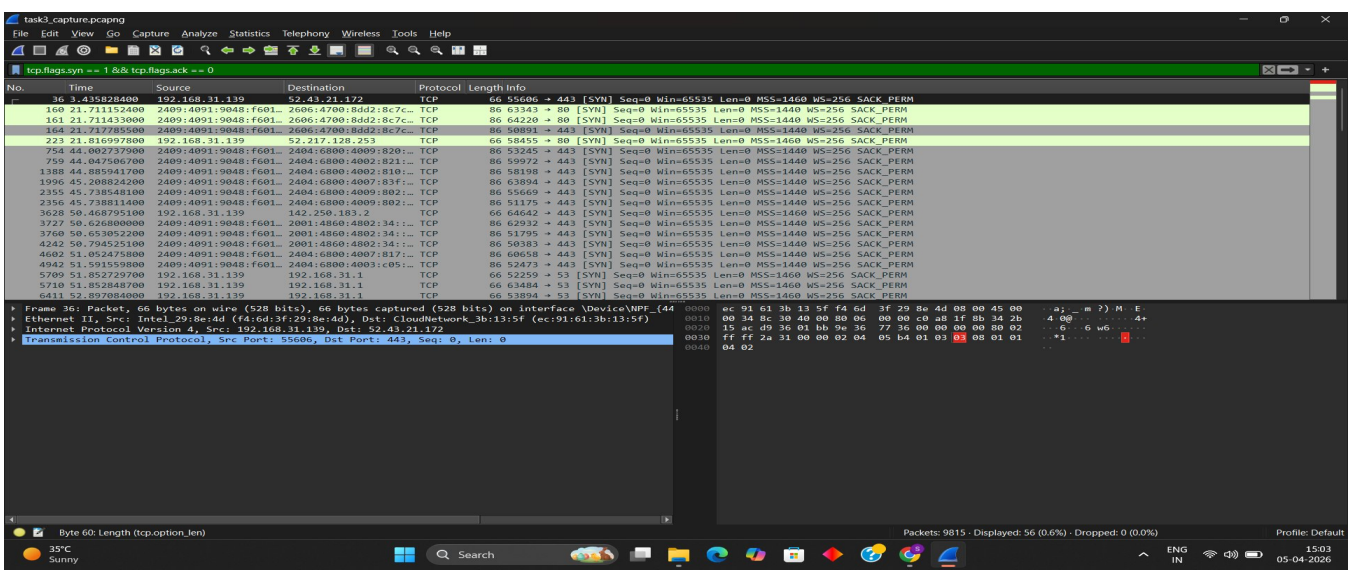
4.1 DNS Traffic (Filter: dns — 290 packets)

DNS queries and responses to google.com, example.com, and googleapis.com. Both A (IPv4) and AAAA (IPv6) record types observed. DNS server: 192.168.31.1.



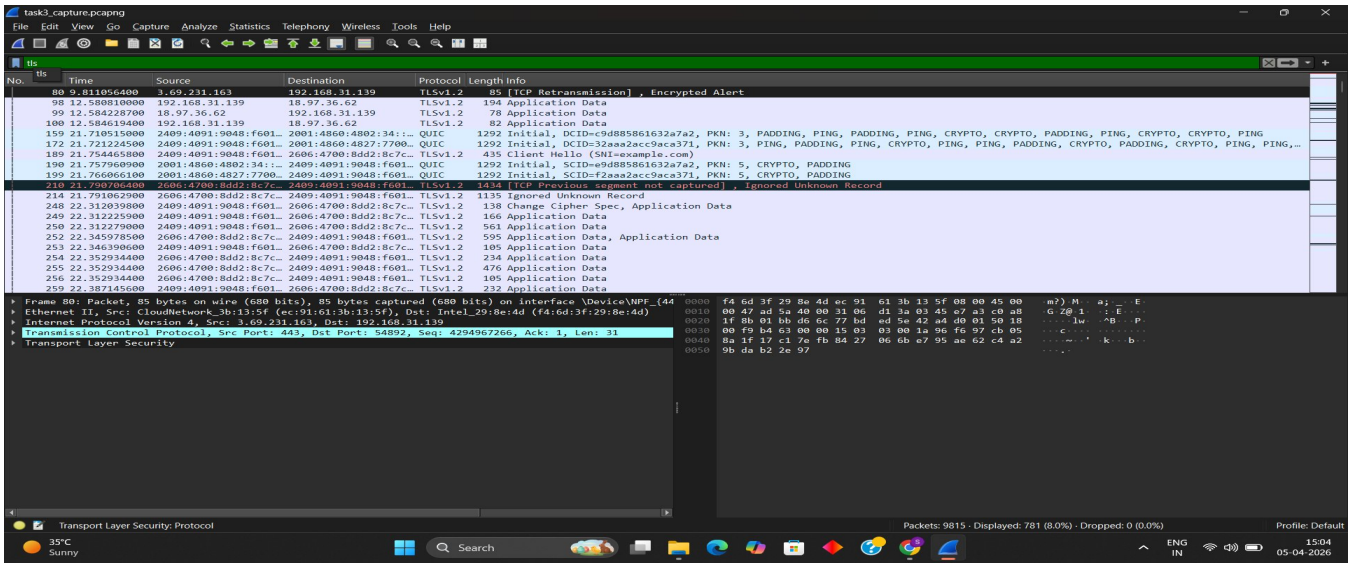
4.2 TCP SYN Handshake (Filter: tcp.flags.syn==1 — 56 packets)

SYN packets initiating TCP connections to port 443 (HTTPS). MSS=1460, WS=256, SACK_PERM flags are standard. No port scanning detected.



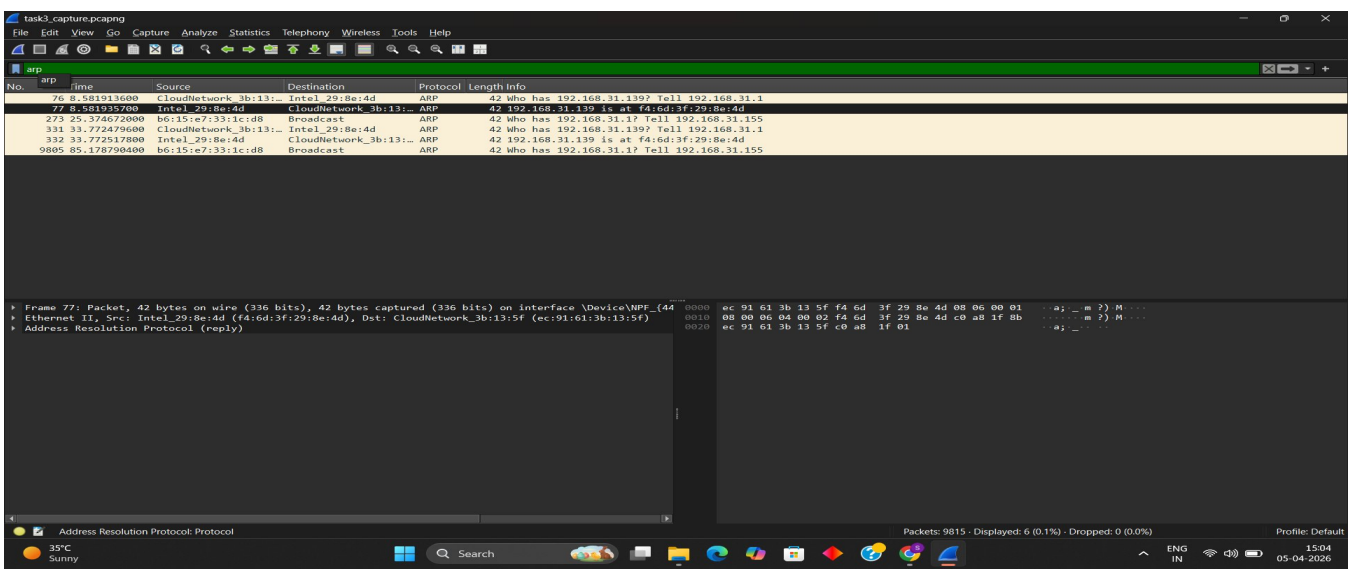
4.3 TLS Encrypted Traffic (Filter: tls — 781 packets)

TLS 1.2 and TLS 1.3 observed. Client Hello with SNI=example.com visible. Application data is encrypted. One TCP Retransmission with Encrypted Alert noted.



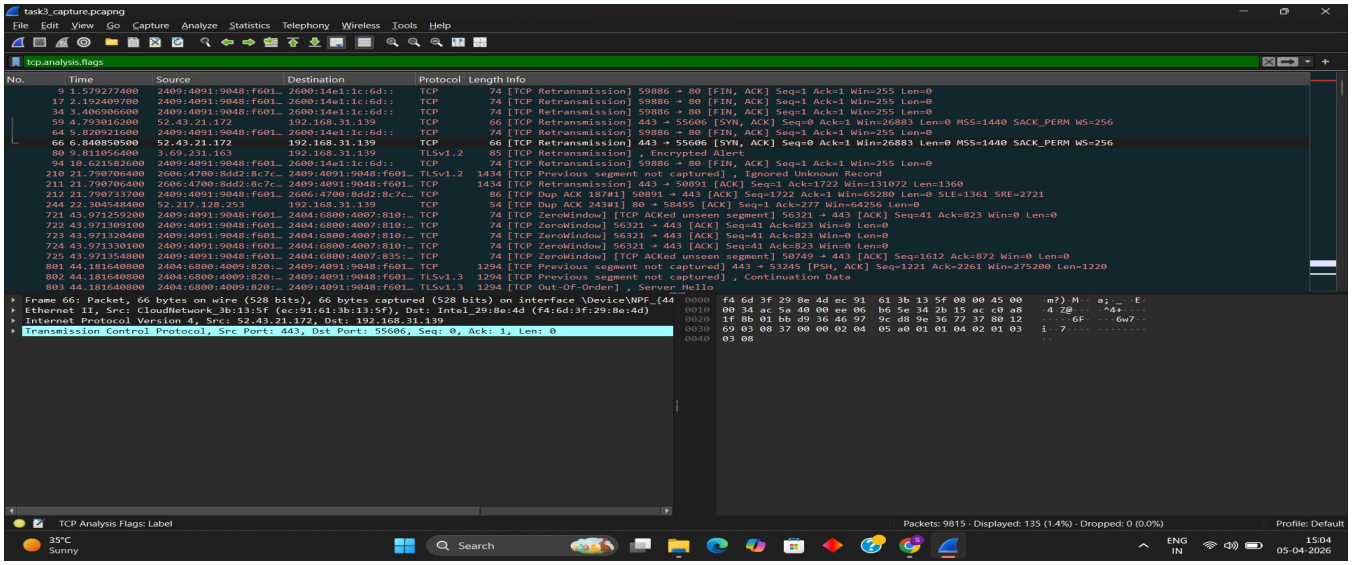
4.4 ARP Packets (Filter: arp — 6 packets)

Normal ARP request/reply for gateway MAC discovery. 192.168.31.139 is at f4:6d:3f:29:8e:4d. No ARP poisoning detected.



4.5 Suspicious / Anomalous Packets (Filter: tcp.analysis.flags — 135 packets)

TCP Retransmissions, ZeroWindow, Out-of-Order, and Dup ACK packets detected. All consistent with normal network conditions — not indicative of an attack.



5. Risk Summary

HIGH	MEDIUM	LOW	INFO
0	0	3	3

6. Detailed Findings

Risk Level	Finding	Description
Low	TCP Retransmissions	Packets 9,17,34,64,66 show FIN/ACK and SYN/ACK retransmits. Indicates minor packet loss — not malicious.
Low	TCP ZeroWindow	Packets 721-725 show ZeroWindow — receiver buffer temporarily full. Normal congestion behavior.
Low	TCP Out-of-Order	Minor packet reordering observed. Common in Wi-Fi networks, not an attack indicator.
Info	ARP Broadcasts	Repeated ARP at packet 9805 (85s) is normal router periodic MAC refresh.
Info	TLS Encrypted Alert	Packet 80 shows Encrypted Alert — a TLS connection closure, not an intrusion.
Info	QUIC Protocol	Google/YouTube QUIC traffic with Protected Payloads — expected modern web traffic.

7. Recommendations

- Enable HTTPS inspection or use a proxy to monitor encrypted TLS traffic for threats.
- Investigate TCP retransmissions — check for unstable Wi-Fi or network congestion.
- Monitor DNS queries regularly to detect DNS tunneling or data exfiltration.
- Set up a ZeroWindow alert rule in Wireshark/IDS for production network monitoring.

- Periodically capture and analyze ARP traffic to detect ARP poisoning attempts.
- Consider enabling Wireshark's built-in protocol dissectors for deeper QUIC analysis.

8. Conclusion

The packet capture and analysis of live Wi-Fi traffic revealed several common protocols in operation including DNS, TCP, TLS 1.2/1.3, ARP, QUIC, and UDP. Minor network anomalies such as TCP retransmissions, ZeroWindow events, and out-of-order packets were detected. These findings are consistent with normal home/office network behavior and do not indicate any active attack or security breach.

No malicious traffic, port scans, DNS tunneling, ARP poisoning, or unauthorized connections were identified in this capture.